

Gianmarco Santoro

Data Scientist - AI | Mechanical Engineer - Automation

Seeking challenges in **Computer Vision, Robotics, Generative AI, LAMs**, and related areas

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DeepFake Detection | Computer Vision - Machine Learning Engineer

Media Integration and Communication Center (MICC) | Florence, Italy | Dec 2023 - Apr 2024

Implemented **Out-of-Distribution (OOD) Detection** on *FaceForensics++* dataset. A 3-step method:

- Extracted Self-Attention heatmaps via **Vision Transformer**, pre-trained on face recognition.
- Exploited heatmaps: trained a **Convolutional Autoencoder** on real ones.
- OOD detection, tested real vs fake images reconstruction error: achieved a **60% AUC-ROC across five forgery types**.

Data Science & Statistical Learning | 2nd Level Master - Postgraduate

UniFi & IMT for Advanced Studies | Feb 2023 - Apr 2024 | 110/110 cum laude

Some topics: **Optimization**, GIS, NLP, Graph Theory. A few projects (visit [Website](#) or [GitHub](#) for more):

- Built **Machine Learning** pipeline with K-Fold Cross-Validation, **Grid Search** (degree hyperparameter) on polynomial regression.
- Developed **Deep Learning** images classifier via **CNNs** models (*PyTorch*), achieving an accuracy of **70.4%** on the test set.
- Predicted salary on selected features using **CART** and other models. Reached **83.8% accuracy** with Gradient Boosting.
- Forecasted fit-class attendance post-booking: achieved **75.8% accuracy** with **logistic regression** on a highly imbalanced dataset.

Smart Center | IT & System Integration Engineer

Daikin Applied Europe SpA | Rome, Italy | Jun 2021 - Sep 2022

Worked on cloud-based HVAC monitoring, system integration, and services:

- **Automated PLC** mapping and alarm selection via **Python**, reducing manual configuration time by **95%**.
- Developed a real-time sensor-to-cloud **temperature control system** for efficiency, well-being, and avoiding manual configuration.
- **Services** such as data aggregation via web API to generate periodic reports for the Service Department.

Wireless Charging System | Operations Research Scientist

Enermove Srl | Turin, Italy | Apr 2020 - Mar 2021

Optimized electric vehicles wireless charging system placement via Genetic Algorithms, improving energy efficiency-to-cost ratio.

Chip Extrusion Recycling | Manufacturing Research Engineer

Institut für Umformtechnik und Leichtbau | Dortmund, Germany | Feb - Sep 2018

Experimented with the **aluminum recycling process** by direct extrusion of mixed waste alloys (DoE). Compared profile properties to simulated (FEM), assessed **industrial feasibility** to identify process bottlenecks, as logistic issues.

Mechanical Engineering | M.Sc. at PoliTo & B.Sc. at UniBo | 104/110

Specialized in **Automation**. Some extra projects: • Automation and **mechatronics** labs: PLC, PID controls, pneumatics, mechanisms.

• Designed a reduction **gearbox**. • Developed a home **heating system**. • Built a **numerical model** for metal flue heat exchange.

Personal Projects | Jun 2022 - Current

- Developed my **Website**. Tools: *HTML, CSS, JS, VScode, GIT & GitHub*, tested across *Arc, Safari, and Chrome* browsers.
- Conducted Market Research, Javelin Board and SWOT analysis, Lean Startup Method on MVPs: Project-Driven Work Community, Food-based Social Media, Selected Products Referral Platform. Tools: *Figma, Mural, Notion, Canva, G-Forms/Sheets/Docs*.
- Sold two houses, closed three related mortgages, and reinvested the proceeds to renovate family-owned bar and house.

Skills | Languages: • Italian: C2 - Mother tongue • English: C1 - Professional • German: A2 - Basic

AI & SW: Python, PyTorch, OpenCV, NumPy, Pandas, Scikit-learn, Matplotlib, Seaborn, GIT-GitHub/Lab, R, Matlab & Simulink

Web, Tools, Industrial: HTML, CSS, Jupyter/Colab, MS-Office (Excel), G-Suite, Notion, Canva, AutoCAD, SolidWorks

Basic: Bash, C, JS, PLC Siemens-Clim., TCP/IP, SSH, mRemoteNG, Modbus, qModMaster, CATIA, HyperMesh/Ext., MSC-Adams/Apex